Final Review: Key Ideas/Techniques

Chapter 6: Review

Key Ideas/Techniques

1. Definition of upper and lower (Darboux) sums.
2. Definition of $\int_a^b f$ in terms of the upper and lower (Darboux) integrals.
3. Cauchy criterion for integrability (definition in terms of ϵ, P).
4. Monotone $\Rightarrow$ Integrable
5. Continuous $\Rightarrow$ Integrable

Activity

Prove, using the Cauchy criterion for integrability (ϵ, P definition) that a real-valued function that is defined and constant on the interval $[a, b]$, except for at some point $x_0 \in (a, b)$, is integrable.

Chapter 5: Review

Key Ideas/Techniques

1. Limit definition of the derivative.
2. Know how to compute $f'(x)$ using the definition.
3. Algebraic properties of the derivative (sum, product, quotient, chain rule)
4. Differentiable $\Rightarrow$ Continuous
5. Mean Value Theorem and applications

Activity

Compute f' if $f(x) = \frac{1}{x}$ using the limit definition.

Activity

Suppose $f, g: \mathbb{R} \to \mathbb{R}$ are differentiable. Prove if $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$, then $f(x) \leq g(x)$ for all $x \geq 0$.

Hint: Consider $h = g - f$.

Activity

Suppose $f: \mathbb{R} \to \mathbb{R}$ is differentiable and $f': \mathbb{R} \to \mathbb{R}$ is also differentiable. Prove if $f(0) = f'(0) = 0$ and $f''(x) \leq M$ for all $x \geq 0$, then $f(x) \leq \frac{M}{2}x^2$ for all $x \geq 0$.

Hint: Apply the previous activity twice.

Activity

Suppose $f\colon (a,b) \to \mathbb{R}$ is differentiable and $f'\colon (a,b) \to \mathbb{R}$ is also differentiable. Prove if $f(x_1) = f(x_2) = f(x_3) = 0$ for some $x_1 < x_2 < x_3$, then $f''(c) = 0$ for some $c \in \mathbb{R}$.

Hint: Apply the MVT to f on the intervals $[x_1, x_2]$ and $[x_2, x_3]$.

Activity

Prove that the equation $x^3 - 3x + 1 = 0$ has exactly three solutions in $\mathbb{R}$.

Hint: Where is the function $f(x) = x^3 - 3x + 1$ increasing/decreasing?

Activity

Suppose $f: \mathbb{R} \to \mathbb{R}$ is differentiable. Prove if

- $|f(x)| \leq 1$ and
- $f'(x) > -1$ for all $x \in \mathbb{R}$,

then the equation $f(x) + x = 0$ has exactly one solution in $\mathbb{R}$.

Activity

Show that the sum of a positive number and its reciprocal is at least 2, i.e. show that $f(x) = x + \frac{1}{x}$ is bounded below by 2 on $(0, \infty)$.

Hint: Determine where f is increasing and decreasing.

Activity

Suppose $f : (0, \infty) \to \mathbb{R}$ is a function such that $f'(x) = \frac{1}{x}$ for all $x > 0$ and $f(1) = 0$. Prove $f(xy) = f(x) + f(y)$ for all $x, y \in (0, \infty)$.

Hint: Fix y and compute the derivative of $f(xy)$ and $f(x) + f(y)$.

Note: $f(x) = \log(x)$, but we haven't studied this function. So don't use it.

Chapter 3: Review

Key Ideas/Techniques – Limits

1. Definition of $\lim_{x \to a} f(x)$ using ϵ, δ and the proof template for verifying it.
2. Equivalent definition of $\lim_{x \to a} f(x)$ using sequences.
3. Algebraic properties of limits (sums, products, quotients)
4. Template for showing $\lim_{x \to a} f(x)$ does not exist
    - Find sequences (x_n) and (y_n) in $\mathrm{dom}(f) - \{a\}$ where $\lim x_n = a = \lim y_n$, but $\lim f(x_n) \neq \lim f(y_n)$.

Activity

Suppose $\mathrm{dom}(f) = \mathbb{R}$ and $\lim_{x \to 3} f(x) = L$. Prove using the ϵ, δ definition that $\lim_{x \to a} f(2x + 1) = L$.

Template: For proving $\lim_{x \to a} f(x) = L$

- "Fix $\epsilon > 0$."
- "Let $\delta = ?$"
- "Assume $0 < |x - a| < \delta$."
- Show $|f(x) - L| < \epsilon$.
- "Since $\epsilon > 0$ was arbitrary, $\lim_{x \to a} f(x) = L$."

Activity

Let

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Prove if $a \in \mathbb{R}$, then $\lim\limits_{x \to a} f(x)$ does not exist.

Template: Find sequences (x_n) and (y_n) in $\mathrm{dom}(f) - \{a\}$ where $\lim x_n = a = \lim y_n$, but $\lim f(x_n) \neq \lim f(y_n)$.

Key Ideas/Techniques – Continuity

1. Definition of continuity using sequences.
2. Equivalent definition using ϵ, δ and the proof template for verifying it.
3. Algebraic properties of continuity (sums, products, quotients, composition)
4. Template for showing a function is not continuous at a point.

Activity

Suppose $f, g\colon S \to \mathbb{R}$ are continuous at x_0. Prove using the ϵ, δ definition that $f + g$ is continuous at x_0.

Activity

Show that

$$f(x) = \begin{cases} \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is not continuous at 0.

Template: Find a sequence (x_n) in $\mathrm{dom}(f)$ where $\lim x_n = 0$, but $\lim f(x_n) \neq f(0)$.

Key Ideas/Techniques – Continuity

1. Maximums and minimums exist for continuous functions on closed intervals.
2. Intermediate Value Theorem.
3. Using the Intermediate Value Theorem in applications.
4. Definition of uniform continuity and the proof template for verifying it.

Recipe for Solving IVT Problems

Theorem. (IVT) Let f be a continuous real-valued function on a closed interval $[a, b]$. If y is a value between $f(a)$ and $f(b)$, then there exists at least one $x \in (a, b)$ such that $f(x) = y$.

I. Define a function $f(x)$.
II. Define a number y.
III. Choose an interval $[a, b]$.
IV. Establish that f is continuous on $[a, b]$.
V. Establish that y is between $f(a)$ and $f(b)$.
VI. Apply the conclusion of the IVT.

Activity

Suppose $f: \mathbb{R} \to \mathbb{R}$ is continuous and $|f(x)| \leq 1$ for all $x \in \mathbb{R}$. Prove that there exists at least one point x_0 such that $f(x) = 6 - x^3$.

Template: for applying the IVT

I. Define a function $f(x)$.

II. Define a number y.

III. Choose an interval $[a, b]$.

IV. Establish that f is continuous on $[a, b]$.

V. Establish that y is between $f(a)$ and $f(b)$.

VI. Apply the conclusion of the IVT.

Activity

Prove that if $a \geq 0$, then the equation $x^2 = a$ has a solution.

Activity

Suppose $f, g: S \to \mathbb{R}$ are uniformly continuous on S. Prove that $f + g$ is uniformly continuous on S.

Template: To show that $f: S \to \mathbb{R}$ is uniformly continuous on S.

I. "Fix $\epsilon > 0$."

II. "Let $\delta = (something)$." [this requires scratch work]

III. "Assume $x, y \in S$ and $|x - y| < \delta$."

IV. Show that $|f(x) - f(y)| < \epsilon$.

V. "Since $\epsilon > 0$ was arbitrary, f is uniformly continuous on S."

Chapter 2: Review

Key Ideas/Techniques – Sequences

1. Definition of $\lim s_n = L$ and the proof template for verifying it.
2. Algebraic properties of limits (sums, products, quotients)
3. Definition of $\lim s_n = \pm\infty$ and the proof template for verifying it.
4. Definition of monotone sequences and limits of monotone sequences
   A. Monotone + Bounded ⇒ Converges to a real number
   B. Monotone + Unbounded ⇒ Diverges to $\pm\infty$.

Activity

Suppose $s_n \geq 0$ for all n and $\lim s_n = s$ with $s > 0$. Prove using the ϵ, δ definition that $\lim \sqrt{s_n} = \sqrt{s}$.

Template: for proving $\lim s_n = s$

I. "Fix $\epsilon > 0$."

II. "Let $N = ?$"

III. "Assume $n > N$."

IV. Show $|s_n - s| < \epsilon$.

V. "Since $\epsilon > 0$ was arbitrary, $\lim s_n = s$."

Activity

Suppose $s_n > 0$ for all n. Prove that $\lim s_n = 0$ if and only if $\lim \frac{1}{s_n} = +\infty$.

Template: for proving $\lim a_n = +\infty$

I. "Fix $M > 0$."

II. "Let $N = ?$"

III. "Assume $n > N$."

IV. Show $a_n > M$.

V. "Since $M > 0$ was arbitrary, $\lim a_n = +\infty$."

Activity

Suppose $a \geq 0$. Prove that

$$\lim a^n = \begin{cases} +\infty, & a > 1 \\ 1, & a = 1 \\ 0, & a < 1 \end{cases}$$

Hint: When $a < 1$ prove that (a^n) is bounded and monotone. Use the previous activity when $a > 1$.

Key Ideas/Techniques – Subsequences

1. Definition of subsequences.
2. The Bolzano-Weierstrass Theorem
3. Definition of the subsequential limit set.
4. Definition of the limit inferior and the limit superior; connection to subsequential limit set.

Activity

a) Construct a sequence with exactly **five** subsequential limits.

b) Construct a sequence whose subsequential limit set is

$\mathbb{N} \cup \{+\infty\}$.

Key Ideas/Techniques – Cauchy Sequences

1. Definition of Cauchy sequences and template for verifying a sequence is Cauchy.
2. Cauchy $\Leftrightarrow$ Convergent

Activity

Let $s_n = \sum_{k=0}^{n} \frac{\sin(k)}{4^k}$. Prove that (s_n) is Cauchy.

Template: for proving (s_n) is Cauchy

I. "Fix $\epsilon > 0$."

II. "Let $N = ?$"

III. "Assume $n, m > N$." [sometimes helpful to also assume $m > n$]

IV. Show $|s_n - s_m| > \epsilon$.

V. "Since $\epsilon > 0$ was arbitrary, (s_n) is Cauchy."

Chapter 1: Review

Key Ideas/Techniques

1. Induction
2. Rational and Irrational Numbers
3. Completeness Axiom, Supremum, Infimum

Activity

Prove by induction: If $a_1, a_2, \ldots$ are odd numbers, then for every $\mathrm{n} \in \mathbb{N}$ the product $a_1 a_2 \cdots a_n$ is also odd.

Activity

Suppose $k > 2$ is irrational and x_1, x_2 are solutions to the equation

$$x^2 + kx + 1 = 0$$

Prove x_1 and x_2 are both irrational.

Hint: What is $x_1 + x_2$ and $x_1 x_2$?

Activity

Compute the maximum, minimum, supremum, and infimum of the following sets:

a) $A = \{x : |x - 3| < 5\} \cap \{x : |x - 2| \geq 4\}$

b) $B = \{x : x^3 < 8\} \cup [7, +\infty)$

c) $C = \{1 + \frac{(-1)^n}{n} : n \in \mathbb{N}$ and $n \geq k\}$ where $k \in \mathbb{N}$